DATE: October 15, 2026

TO: Dr. Haris Vikalo
Mugdha Jadao

FROM: Ethan Chua
Esteban Martinez
Li Shi
Andrew Yin
Abhirit Das

SUBJECT: Project Definition for BevoTag, an Open Asset-Tracking Tag for the UT Austin Campus

INTRODUCTION

This Project Definition defines the design problem that the BevoTag team will solve in Senior Design and explains how the five-member team has organized itself to solve it. BevoTag is a small tag that a student or staff member attaches to a laptop, bag, or piece of university equipment to locate it anywhere on the University of Texas at Austin campus. Satellite positioning such as GPS fails inside buildings, and consumer tags such as the Apple AirTag update an item's location only when a nearby phone from the same manufacturer detects them [1]. BevoTag is open in the sense that it needs no manufacturer's phone network: it estimates the item's position from the Wi-Fi networks already installed across campus and from its own motion sensor, and it reports that position to its owner on a campus map. In short, the team will build a tag, firmware, a building survey, and an application that together place an item within a median of 5 m and in the correct room in more than 90% of trials.

The design problem is to track an item continuously, indoors and outdoors, without new infrastructure. Wi-Fi measurements alone can mislocate a device by several meters because signals fluctuate as the environment changes, and a low-cost motion sensor alone drifts by tens of meters within a minute; combining the two lets each correct the other. The project will test this approach in surveyed areas beginning with the Engineering Education and Research Center (EER). Wi-Fi, radio-emission, data-security, and battery-safety standards constrain the design, as does university network policy. The team commits to recording no one's network traffic, to encrypting and limiting location data, and to using the campus network only as an ordinary client.

The requirements turn this need into measurable targets. Beyond the accuracy targets, the system must update an item's location at least every 10 s while it moves and run at least 24 h on one battery charge. Two sub-teams will meet these targets: Esteban Martinez and Li Shi will build the tag hardware and firmware, and Andrew Yin and Abhirit Das will build the localization software and the user application, with Ethan Chua coordinating as Project Lead. The team communicates on Discord, tracks every task on GitHub, reviews all code and writing before submission, and decides contested questions by majority vote after testing the options.

PROBLEM STATEMENT

BevoTag must locate a tagged item to within a few meters across indoor and outdoor areas of campus while relying only on existing campus Wi-Fi and its own motion sensor. Buildings block and reflect satellite signals [2], and low-cost motion sensors drift [3], so no single source of position is reliable on its own. The Design Problem Description shows that fusing Wi-Fi fingerprints with motion sensing meets this need within the limits that standards and ethical obligations set. The Requirements Specification then turns that need into measurable targets, led by a median position error under 5 m.

Design Problem Description

A successful BevoTag design must estimate an item's location indoors and outdoors without infrastructure the team cannot control, and it must show that location to the item's owner on a map. Existing trackers fail this need because satellite signals do not reach indoors and crowdsourced tags depend on passing phones. BevoTag addresses the need by fusing Wi-Fi fingerprints with motion sensing, an approach that brings three main challenges: limited power on the tag, radio signals that change from moment to moment, and sensor drift. Because the tag reports location, the design must also satisfy radio, data-security, and battery-safety standards and protect the privacy of bystanders and owners.

The Need for On-Campus Item Tracking

Students, faculty, and staff need a way to find misplaced items anywhere on the main UT Austin campus, known as the Forty Acres. Laptops, bags, and shared university equipment move between classrooms, labs, libraries, and outdoor plazas many times a day, and an item left behind

can be anywhere among dozens of multistory buildings. A useful tracker must therefore work indoors, on any floor, and in the narrow outdoor spaces between tall buildings. The team's goal is tracking that needs neither new infrastructure, such as beacons installed in every building, nor a passerby who owns a particular brand of smartphone.

Why Existing Solutions Fall Short

Three existing approaches to tracking fail on campus for different reasons. Satellite positioning cannot reach indoors, crowdsourced Bluetooth tags stop updating when no compatible phone passes by, and Wi-Fi positioning based on signal strength alone errs by enough to place an item in the wrong room.

Satellite positioning fails indoors and degrades between tall buildings. Global Navigation Satellite Systems (GNSS), the family of satellite positioning systems that includes GPS, compute a receiver's position from faint signals broadcast by satellites about 20,200 km above the Earth [4]. Walls and roofs block those signals. Between tall buildings, signal blockage and multipath, the arrival of signals only after they reflect off nearby walls, degrade GNSS so much that earlier low-cost urban systems reported position errors greater than 9 m [2, Sec. 1.1.1].

Crowdsourced tags locate an item only when a compatible phone happens to pass nearby. The Apple AirTag and the Samsung Galaxy SmartTag broadcast a short-range Bluetooth signal, and nearby phones from the same manufacturer detect that signal and report its location [1], [5]. In an empty lab at night no such phone passes by, so the tag's location stops updating until one does.

Wi-Fi positioning based only on signal strength is too coarse for room-level tracking. These systems estimate position from the received signal strength indicator (RSSI), a measure of how strong each nearby Wi-Fi signal is when it reaches the device. Signal strength fluctuates as the environment changes [6], so RSSI fingerprinting typically locates a device only to within 1 to 5 m [7, Table 4], which can place it in the wrong room.

Theory of Operation

BevoTag combines two sources of position information whose errors differ: campus Wi-Fi measurements, which are noisy but do not drift, and a motion sensor, which is smooth over short periods but drifts. A fusion algorithm uses each source to correct the other. The final design may compute this estimate on the tag or on a server; the team will decide after measuring the prototype hardware, as the Design Challenges explain.

Wi-Fi measurements locate the tag by matching them against a survey of each building. UT Austin operates campus Wi-Fi networks, including utexas-iot, which serves Internet of Things devices once their owners register each device's hardware address [8]. In addition to RSSI, some Wi-Fi chips report channel state information (CSI), the frequency response of each of dozens of narrow sub-channels, which describes how much the signal on each sub-channel is weakened and delayed along its path [9]. Walls, furniture, and room geometry shape these values, so CSI acts as a detailed fingerprint of a location. The team will first survey a building by recording Wi-Fi measurements at many points whose positions it knows. The system can then estimate the tag's position by finding the survey points whose measurements best match the tag's current measurements.

A motion sensor tracks the tag between Wi-Fi fixes but drifts when used alone. An inertial measurement unit (IMU), similar to the sensor that rotates a smartphone's screen, measures acceleration along three axes and rotation about them. Adding up these measurements from a known starting point, a technique called dead reckoning, tracks the tag's movement many times per second. Small sensor errors accumulate, however, so the estimate drifts farther from the truth the longer it runs: a simple low-cost inertial system can accumulate more than 150 m of average position error within 60 s [3].

A fusion algorithm combines the two sources so that the result neither jumps like Wi-Fi alone nor drifts like the IMU alone. A standard candidate is the extended Kalman filter (EKF) [10], which uses the IMU to predict how the tag has moved and uses each new Wi-Fi fix to correct the accumulated drift, weighting each source by how reliable it currently is. Outdoors, where satellite signals are available, the same algorithm can also accept a GNSS position as an optional third

source.

Design Functionality and Scope

BevoTag is a system of a tag, a location server, and a user application that together turn Wi-Fi and motion measurements into a position on a campus map. The project covers a working prototype tested in surveyed areas starting with EER. It excludes cellular connectivity, a campus-wide survey, mass-production tooling, and the tracking of people.

Figure 1 on the next page shows how the parts of the system work together. The tag measures nearby Wi-Fi signals and its own motion (step 1) and sends these measurements over the campus Wi-Fi network to the location server (step 2). The server combines them with a previously recorded survey of the building to estimate the tag's position, or it receives an estimate that the tag computed itself (step 3). The user application then displays that position on a campus map (step 4). To deliver this behavior, any BevoTag design must perform four functions:

- **Wi-Fi measurement:** record the RSSI and, where the hardware supports it, the CSI of signals from nearby campus access points in the 2.4 GHz band.
- **Motion tracking:** sample the IMU's acceleration and rotation fast enough to follow a walking person carrying the tag.
- **Position estimation:** fuse the motion data with Wi-Fi position fixes, correcting the drift that accumulates in the motion estimate.
- **Reporting:** send the latest position and device status to an application that the owner can view.

The project includes a working prototype tag, its firmware, a Wi-Fi survey of designated test areas, and a user application. The tag will combine a Wi-Fi chip that can report CSI, such as the ESP32-S3 [9], with a low-noise IMU on a development board or a custom circuit board. Its firmware will read the sensors and either process the measurements or send them to the server. The survey will begin with EER and its adjacent outdoor plazas, and the application will display the tag's position on a campus map.

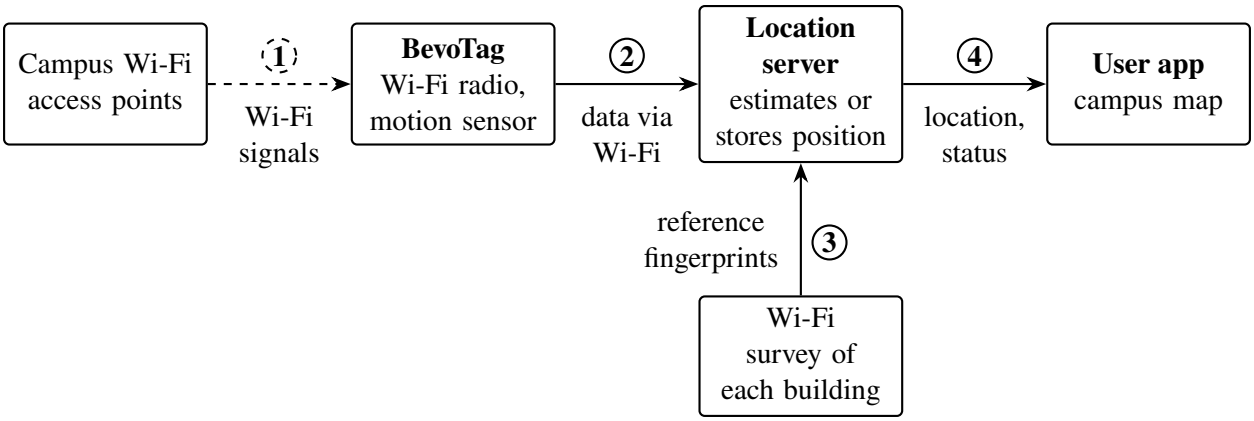


Figure 1. How BevoTag Locates a Tagged Item

Four related capabilities fall outside the project's scope. The tag will not use a cellular connection such as LTE-M or NB-IoT, because these require a paid data plan and a second radio that would raise cost and power consumption, and campus Wi-Fi already covers the target areas. The team will not survey every building on campus, because a survey requires walking each floor and recording measurements at many points. It will instead test in a few defined benchmark areas and document the survey method so that others can add buildings later. The team will 3D-print or buy the prototype enclosure rather than design it for injection molding, since the goal is a working proof of concept rather than a manufacturable product. Finally, BevoTag locates objects, not people, and the design excludes features meant to follow a person, such as continuous covert tracking.

Design Challenges

Three challenges will most strongly shape the design: deciding where to compute the position, coping with radio signals that never repeat exactly, and correcting motion-sensor drift quickly enough. Each challenge requires a measurement on prototype hardware before the team can finalize the design.

The first challenge is deciding whether to compute the position on the tag or on the server. Processing CSI and running a fusion filter on a small microcontroller strains its memory and battery. Sending raw measurements to the server moves that work off the tag but keeps the radio on longer, which also drains the battery. The team will settle this trade-off by measuring CSI capture rates, IMU data rates, and power draw on the prototype.

The second challenge is that indoor radio signals vary from moment to moment. The Wi-Fi measurements recorded at a given spot fluctuate with noise and with changes in the environment, such as people walking through hallways [6], so the matching method must tolerate measurements that never exactly repeat the survey. The team will therefore record survey measurements at several times of day and test the matching method on data that it did not use to build the survey.

The third challenge is correcting IMU drift before it grows large. Low-cost motion sensors accumulate tens of meters of position error within a minute [3], so Wi-Fi fixes must arrive often enough, and be reliable enough, to keep the fused estimate from diverging from the true position. This requirement ties the Wi-Fi update rate directly to the accuracy targets in the Performance Specifications.

Relevant Standards

Six standards and policies constrain how BevoTag uses the radio spectrum, connects its sensors, protects its data, powers itself, and uses the campus network. Their most direct effects are that the tag must use a Wi-Fi chip that exposes CSI, a pre-certified radio module, encrypted telemetry, and a certified, protected lithium battery. Appendix A lists the specific clauses of each standard, how each affects the design, and how the team will verify compliance.

IEEE 802.11 [11] limits BevoTag to Wi-Fi chips that expose CSI and governs how the tag joins campus networks. The standard defines how Wi-Fi devices format and exchange signals, including the training fields at the start of every packet from which a receiver derives CSI [9].

Federal Communications Commission (FCC) rules in Title 47 of the Code of Federal Regulations, Part 15 [12], lead the team to use a Wi-Fi module that already holds FCC certification. Part 15 limits both the intended Wi-Fi transmissions and the unintended electrical noise that the tag's processor and circuits emit, so that the tag does not interfere with campus communications. A module that already holds a modular grant can be used in a new product under its existing FCC ID [12, §15.212], which makes it the most practical way to meet these limits.

IEEE 1451.0 [13], the base standard of the IEEE 1451 smart transducer family, leads the team to separate sensor drivers from the position-estimation code. The standard describes how sensors should present their data and calibration information to a processor. Following its approach lets the team replace either sensor without rewriting the fusion algorithm.

NIST Special Publication 800-88 [14] and FIPS 140-3 [15] determine how the tag encrypts its reports and how a factory reset erases its memory. These federal guidelines cover the erasure of stored data and the cryptographic methods that protect data, and they apply because the tag stores network credentials and location history.

IEC 62133-2 [16] sets the safety requirements for the rechargeable lithium battery that powers the tag. Its tests address overcharge, short circuit, and overheating, so the team will choose a cell certified to the standard and add a protection circuit on its own board.

The university's acceptable use policy [17] governs how the tag may use the campus network. It forbids scanning or probing any information resource without authorization and interfering with other users' access, which reinforces the ethical commitments below.

Ethical Considerations

Because BevoTag's purpose is to report location, it creates obligations to people beyond its owner. The team commits to five measures: recording only the signal measurements needed for positioning, protecting stored location data, never disrupting the campus network, protecting users from battery hazards and electronic waste, and honoring the licenses of the software and map data the project uses.

The firmware will record only signal measurements and packet header fields, never the content of anyone's network traffic. To capture CSI, the tag listens to Wi-Fi traffic that may come from other people's devices. It will keep only the fields needed for positioning, such as the access point address and signal strength, and it will not decode, store, or transmit the contents of any packet.

The team will treat location history as personal information, because a history of where an item

has been reveals where its owner has been. The team will encrypt data in transit between the tag and the server and restrict server access to the authenticated owner of each tag. It will also delete stored location history at the end of the project and ensure that a factory reset erases stored credentials and history, consistent with the guidelines in Appendix A.

The tag will use the campus network only as an ordinary client. It will listen passively or connect through networks, such as utexas-iot, that the university provides for devices of this kind [8]. It will never transmit traffic designed to disrupt, congest, or probe UT Austin's network.

The design will protect people from battery hazards and limit electronic waste. The tag carries a small lithium battery near people and their belongings, so it will include protection against overcharging, short circuits, and overheating. To limit waste, the team will favor a rechargeable battery over disposable cells and will document how to dispose of the prototype's battery properly.

The team will respect the intellectual property of others. The firmware, server, and application will build on third-party libraries and map data whose licenses may require attribution or limit redistribution. The team will record the license of each component it uses, comply with those terms, and cite the published research behind its methods in every project document.

Project Deliverables

At the end of EE 464, the team will deliver to Dr. Vikalo an integrated prototype that estimates and displays the location of a tagged item within the surveyed campus areas, together with the data and documentation needed to extend it. The delivery will consist of six items:

- **BevoTag hardware device:** a compact, battery-powered tag in an enclosure that collects Wi-Fi and motion measurements and sends them to the location server.
- **Embedded firmware:** software on the tag that reads the sensors, timestamps each measurement, monitors device status, and communicates over Wi-Fi.
- **Campus survey and map:** Wi-Fi survey data and a digital 3D map of the designated test areas that together let the system estimate, display, and evaluate the tag's location.

- **Localization backend:** server software that receives data from each tag, produces or stores its position estimate, and provides the result to the user application.
- **User application:** a web or mobile application in which a user can view a tag's estimated location and status on the campus map.
- **Documentation:** the source code, the survey procedure, and test results measured against the specifications in this report.

Requirements Specification

A successful BevoTag implementation must locate a tagged item with a median error under 5 m, identify its room in more than 90% of trials, update its location at least every 10 s while it moves, and run at least 24 h on one battery charge. The tables in this subsection set these and the other measurable targets for the system's inputs and outputs, user interface, operating environment, and performance, without prescribing an implementation. Wi-Fi and inertial sensing are the primary inputs, and GNSS may supplement them outdoors. The team set these targets from published Wi-Fi positioning results [7] and its own judgment of what makes a tracker useful, and it will confirm them with Dr. Vikalo before detailed design.

Input and Output Specifications

The BevoTag system has three components, the tag, a backend server, and the user application, and its primary inputs are Wi-Fi signals and the tag's own motion. Its outputs are the tag's estimated position, the time of that estimate, and the tag's status, which the application shows to the user. Figure 2 on the next page shows how these inputs and outputs connect, and Table 1, which follows it, sets their values.

Figure 2 shows that every measurement enters through the tag and every output leaves through the application. The tag measures Wi-Fi signals, motion, and, optionally, GNSS signals, and it sends these measurements and its status to the server as telemetry. The server stores the building survey and map, produces or receives the position estimate, and exchanges location, status, and tag selections with the application. A USB-C connection charges the tag and lets a computer running the application configure and debug it.

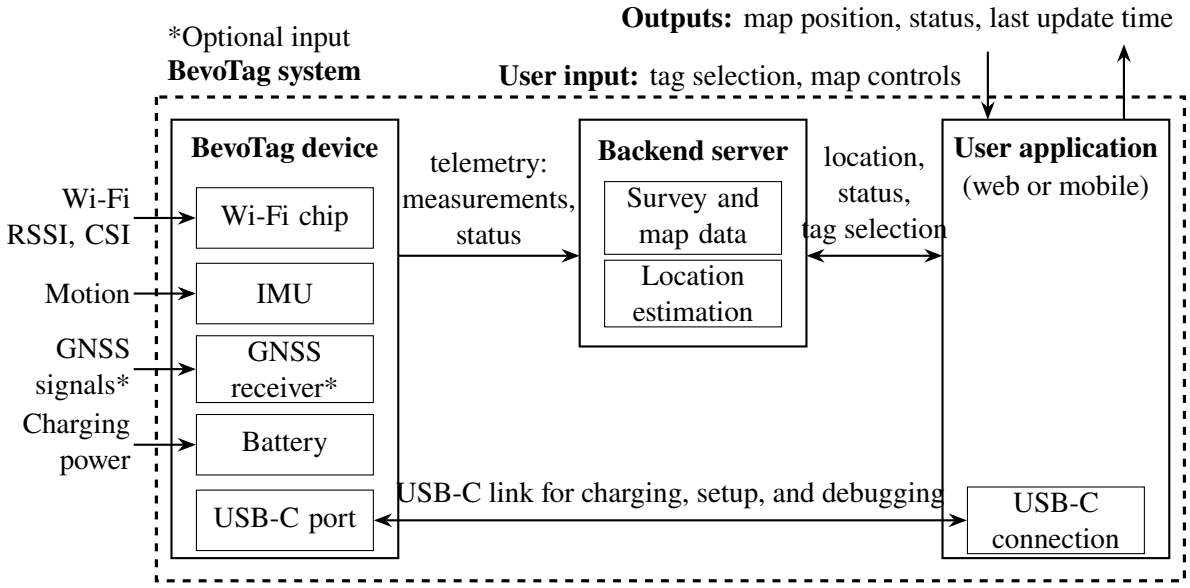


Figure 2. BevoTag Input and Output Diagram

Table 1 lists each input and output in Figure 2 with its expected range and units. The Wi-Fi chip must measure signals as weak as −100 dBm, and the tag must send telemetry at least every 10 s while it moves, so that the application can show each new estimate within 1 s of its arrival. GNSS appears only as an optional input for open outdoor areas.

Table 1. BevoTag Input and Output Specifications

Input/Output	Description	Value or Range
Wi-Fi RSSI input	Strength of each Wi-Fi signal	−100 to −30 dBm
Wi-Fi CSI input	Amplitude and phase per sub-channel [9]	Up to 64 sub-channels
Acceleration input	Linear acceleration from the IMU	±8 g
Angular rate input	Rotation rate from the IMU	±1000 degrees/s
GNSS input (optional)	Satellite position outdoors	Latitude and longitude
Charging power input	Power that charges the battery	5 V through USB-C
User input	Tag selection and map controls	Up to 5 tags per user
Telemetry	Data sent from tag to server	At most 10 s between reports
Location output	Estimated position of the tag	Latitude, longitude, and floor
Timestamp output	Time of the latest estimate	Resolution of 1 s
Device status output	Battery level and connectivity	0 to 100% charge; online or offline
Display refresh output	Time to show a new estimate	At most 1 s

User Interface Specifications

The user application must let a user select a tag, see its estimated location and the time of that estimate on a 3D campus map, and check its status, and it must accept a direct USB-C connection to a tag for configuration and debugging. Table 2 sets a value for each of these requirements. The most important is the location display, which must show each new estimate within 1 s of its arrival. The initial implementation may be either a web or a mobile application, provided that it meets every requirement in the table.

Table 2. User Interface Specifications

Interface	Requirement	Value or Criterion
Device location	Show the tag on a 3D campus map	Building, floor, and position, within 1 s of each estimate
Device selection	Select the tag to monitor	Up to 5 tags per user
Last update time	Show when the position was estimated	Date and time to 1 s
Device status	Show battery level and connectivity	Battery percentage; online or offline
Map interaction	Pan, zoom, and change floors	Response within 1 s of each input
USB-C connection	Configure and debug a connected tag	Tag recognized within 5 s

Operating Environment Specifications

BevoTag must operate in the physical conditions of the surveyed campus zones and within the computing and network environment that the university and the team provide, and Table 3 on the next page sets both. The test zones define the physical conditions: walls and floors that block signals, pedestrian traffic, temperatures from 0 to 40 °C, and a carried tag that changes orientation and moves at up to 5 m/s. The computing and network environment matters because most of BevoTag's function is software. The tag must fit within a microcontroller's memory and join the utexas-iot network, and its telemetry must be encrypted in transit. The backend must run on a Linux server reachable from campus, and the application must run in current browsers or on current phones.

Table 3. Operating Environment Specifications

Condition	Description	Value or Criterion
Physical environment		
Deployment area	Surveyed indoor and outdoor zones, beginning with EER	Surveyed test zones only
Radio environment	Signals arrive directly or only after reflection or obstruction	Line of sight and blocked, indoors and outdoors
Temperature	Ambient temperature for correct operation	0 to 40 °C
User motion	Speed of the person carrying the tag	0 to 5 m/s
Computing and network environment		
Tag processor	Battery-powered microcontroller with integrated Wi-Fi	At most 512 KB of on-chip RAM for firmware
Tag network	Campus network that carries telemetry	utexas-iot at 2.4 GHz, WPA2 with AES, device registered by hardware address [8]
Data transport	Protection of telemetry between tag and server	TLS 1.2 or later
Backend host	Server that receives telemetry and serves the application	Linux virtual machine reachable from campus, with at least 2 CPU cores and 4 GB of memory
Application platform	Devices that run the user application	Current Chrome, Firefox, and Safari browsers, or iOS 17 and Android 13 and later

Performance Specifications

Table 4 on the next page sets the targets that define useful tracking: accuracy, timeliness, battery life, and size and cost. Published Wi-Fi positioning systems show that meter-level accuracy is achievable but that results vary widely with the sensing method and the radio environment [7], so the team defines every target as a repeatable measurement. The team will measure accuracy at surveyed points whose true positions it has recorded independently of BevoTag's estimate, in stationary and moving trials indoors and outdoors. It will repeat the trials at different times of day to capture changing conditions such as pedestrian traffic.

Table 4. Performance Specifications

Parameter	Description	Value or Range	Units
Median position error	Median horizontal distance between estimated and true position during trials	< 5	m
Room identification	Fraction of trials in which the tag is placed in the correct surveyed room or region	> 90	%
Floor identification	Fraction of indoor trials in which the tag is placed on the correct floor	> 95	%
Update interval	Maximum time between location estimates while the tag moves	≤ 10	s
End-to-end latency	Maximum time from a location estimate to its display in the application	< 5	s
Time to first fix	Maximum time from startup to the first usable position estimate	< 30	s
IMU sampling rate	Accelerometer and gyroscope sampling rate during active tracking	≥ 100	Hz
Battery life	Operating time on one charge at one location report per minute	≥ 24	h
Tag size	Largest dimensions of the tag in its enclosure	$\leq 80 \times 50 \times 25$	mm
Tag mass	Mass of the tag with battery and enclosure	≤ 100	g
Parts cost	Cost of components for one prototype tag	≤ 50	US$

TEAM ORGANIZATION

The five-member team has organized itself into two sub-teams under a Project Lead: Embedded Hardware and Firmware, which builds the tag, and Localization and Application Software, which turns measurements into a location and shows it to the user. Each member's assignment matches that member's coursework and experience, and all members share integration, testing, and documentation. The team coordinates through Discord and GitHub, finishes work 48 hours before each course deadline, reviews all work before submission, and resolves disputes with evidence and a majority vote.

Project Roles

The team divided its work along the boundary between the tag and the software that interprets the tag's data. The hardware and firmware sub-team owns everything that collects measurements, and the software sub-team owns everything that turns measurements into a displayed location. The Project Lead and the Documentation Lead coordinate the work that crosses that boundary.

Sub-Teams

The Embedded Hardware and Firmware sub-team, Esteban Martinez and Li Shi, is responsible for the tag itself. It will first build a data logger for feasibility experiments and Wi-Fi survey collection, then integrate the required processing on the tag and evaluate measurement reliability, execution time, memory use, and power consumption. The Localization and Application Software sub-team, Andrew Yin and Abhirit Das, is responsible for turning measurements into a location and presenting it to the user. Its work spans survey data processing, the position-estimation pipeline, the backend that receives and manages tag data, and the user application. The two sub-teams will integrate the tag, the positioning software, and the application together, and all members will take part in system testing.

Personnel and Team-Wide Roles

Each member's assignment draws on specific coursework and experience, and Table 5 on the next page summarizes these assignments. Two members also hold team-wide roles. Ethan Chua serves as Project Lead, and Andrew Yin serves as Documentation Lead, responsible for checking every submission against the ECE Style Guide before the team submits it.

Ethan Chua brings positioning research experience from the UT Radionavigation Laboratory, project experience from Texas Aerial Robotics, and signal processing preparation from ECE 351M, Digital Signal Processing. As Project Lead, Ethan will coordinate milestones, track progress, lead integration testing, and serve as the primary contact with Dr. Vikalo and the Technical TA.

Esteban Martinez and Li Shi have completed ECE 445L, Embedded Systems Design Lab; ECE 422C, Software Design and Implementation II; and ECE 360C, Algorithms. Both have

embedded systems experience through Texas Aerial Robotics and industry internships. Their assignments divide the tag between them. Esteban will lead component selection, power-system design, hardware integration, and electrical testing, and Li will lead firmware development, including Wi-Fi CSI and RSSI capture, IMU sampling, measurement timestamping, and network communication.

Andrew Yin and Abhirit Das have completed the same three courses, and each brings application experience that matches one half of the software sub-team's work. Andrew's experience in mobile application development and machine learning supports Andrew's work on Wi-Fi survey processing, position estimation, sensor fusion, and backend development. Abhirit's experience in web application development and in profiling machine-learning deployment tools at NXP Semiconductors prepares Abhirit to develop and test the user application.

Table 5. Project Responsibilities

Member	Sub-Team or Role	Planned Contributions	Areas of Expertise	Related Coursework
Ethan Chua	Project Lead	Coordination, integration, verification	RF positioning, GNSS, signal processing	ECE 351M
Esteban Martinez	Embedded Hardware and Firmware	Components, power system, hardware integration	Embedded hardware design	ECE 445L, 422C, 360C
Li Shi	Embedded Hardware and Firmware	Firmware, sensor drivers, Wi-Fi communication	Embedded firmware design	ECE 445L, 422C, 360C
Andrew Yin	Localization and Application Software; Documentation Lead	Survey processing, localization, backend	Mobile development, machine learning	ECE 445L, 422C, 360C
Abhirit Das	Localization and Application Software	User application, map and status display	Web development, machine-learning deployment	ECE 445L, 422C, 360C

Team Policies

The team's policies set how members communicate, what they owe one another, and how the team decides questions and resolves conflict. Members use Discord for daily communication and GitHub for code and task tracking, attend two team meetings each week and a mentor meeting every two weeks, and finish work 48 hours before each course deadline. They review one another's work and settle disagreements with evidence and a majority vote. The team will revisit these policies at the end of the semester as it prepares for detailed design.

Project Management Tools

The team coordinates through four shared platforms, each with a defined purpose. Discord is the primary communication channel, with separate channels for general coordination, each sub-team, and meeting logistics so that discussions stay organized and searchable. A shared Google Calendar holds meeting times, course deadlines, and internal milestones. A shared GitHub repository keeps firmware, application code, data-processing scripts, and circuit board and enclosure designs under version control, and GitHub Issues and a GitHub Projects board record each task's owner, due date, and status. A shared Google Drive folder stores reports, meeting notes, and survey data. The Project Lead handles email with Dr. Vikalo and the Technical TA and copies the full team, so every member receives the same guidance.

Team Expectations

Members expect one another to communicate promptly, attend meetings prepared, review one another's work, and finish early. These expectations give every member the same standard for accountability.

Members answer direct questions on Discord within 24 hours on weekdays and flag time-sensitive issues, such as a blocked task or an approaching deadline, by mentioning the affected members directly. The team meets twice a week and meets with Dr. Vikalo every two weeks. Members attend every meeting having reviewed the agenda and notify the team on Discord at least 24 hours before an unavoidable absence. Members who miss a meeting read the notes and follow up on the action items assigned to them.

Every deliverable must meet the standard that the team would present to its mentor. A second member reviews every pull request before it is merged, checks every circuit board design before the team orders boards, and proofreads every written section against the ECE Style Guide. At the start of each meeting, members report what they completed, what they plan to do next, and any blockers, and the Project Lead records these updates and keeps the Projects board current.

Internal deadlines fall at least 48 hours before each course deadline so that the team has time to integrate, review, and revise its work. Members who expect to miss an internal deadline tell the team at least 48 hours in advance, so that the team can re-scope or redistribute the work.

Conflict Resolution

The team will make decisions by vote after research, settle technical disputes with tests, mediate personal conflicts, and respond to missed responsibilities in escalating steps. Each process ends with a clear point of escalation to the Technical TA or the course instructor.

The team makes significant decisions, such as selecting a positioning approach, choosing components, or changing the project scope, by majority vote after researching the options. The members most familiar with the question present the trade-offs of each option, supported by datasheets, published results, cost estimates, or the team's own measurements, and the Project Lead records the outcome and its rationale in the meeting notes. Each sub-team makes decisions that affect only its own work and reports them at the next meeting. When the two members of a sub-team disagree, the Project Lead breaks the tie.

The team settles technical disagreements with data. Where practical, it runs a quick test or trade study against the performance specifications in Table 4 before voting. It brings any unresolved disagreement that poses significant risk to the project to Dr. Vikalo or the Technical TA.

Members first address interpersonal conflicts privately with one another. If that conversation does not resolve the issue, an uninvolved member mediates. The team escalates any conflict that it cannot resolve internally to the Technical TA or the course instructor.

The team responds to missed responsibilities in escalating steps. The Project Lead first raises a missed commitment privately with the member to understand the cause and agree on a recovery plan. The team discusses repeated missed deadlines or meetings at a team meeting and records the corrective actions in the notes. If the problem continues, the team will notify the Technical TA and the course instructor and will report each member's contributions accurately in peer evaluations.

CONCLUSION

This Project Definition establishes what the BevoTag team must build and how the team will work together to build it. Satellite positioning fails inside buildings, and existing tags depend on other people's phones, so the team will build a low-cost tag that locates an item on campus by matching Wi-Fi measurements against a building survey and fusing them with motion-sensor data. The design must place an item within a median of 5 m and in the correct room in more than 90% of trials, within the limits that radio, data-security, and battery standards and the team's ethical commitments set. Two sub-teams under a Project Lead will carry out the work.

The problem matters beyond lost laptops, because reliable positioning where satellite signals fail underlies many location-aware services, and the project's outcome depends on three risks. The tag must capture CSI on low-power hardware, room-level accuracy depends on careful building surveys, and campus network coverage and access policies may limit where the tag can operate. If CSI capture proves unreliable, signal-strength fingerprinting fused with motion data remains a fallback at reduced accuracy. With that fallback and the team's experience in embedded systems, positioning research, and software development, a working prototype within a few surveyed buildings is a realistic outcome.

Next, the team will confirm the specification targets with Dr. Vikalo, complete a data logger for initial Wi-Fi and IMU measurements, and survey EER as the first test building. It will then refine the system block diagram and component selection, and each member will plan an individual risk reduction activity in that member's area of the design.

REFERENCES

[1] Apple Inc. (2026). *AirTag* [Online]. Available: https://www.apple.com/airtag/.

[2] J. E. Yoder and T. E. Humphreys, "Low-cost inertial aiding for deep-urban tightly coupled multi-antenna precise GNSS," *NAVIGATION*, vol. 70, no. 1, art. no. navi.561, Mar. 2023, doi: 10.33012/navi.561.

[3] O. J. Woodman, "An introduction to inertial navigation," Comput. Lab., Univ. Cambridge, Cambridge, U.K., Tech. Rep. UCAM-CL-TR-696, Aug. 2007, doi: 10.48456/tr-696.

[4] National Coordination Office for Space-Based Positioning, Navigation, and Timing. (n.d.). *Space Segment* [Online]. Available: https://www.gps.gov/space-segment.

[5] Samsung Electronics Canada. (n.d.). *Galaxy SmartTag2* [Online]. Available: https://www.samsung.com/ca/business/mobile-accessories/galaxy-smarttag2-white-ei-t5600bwegca/.

[6] S. He and S.-H. G. Chan, "Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 466–490, 2016, doi: 10.1109/COMST.2015.2464084.

[7] A. S. Yaro, F. Maly, and P. Prazak, "A survey of the performance-limiting factors of a 2-dimensional RSS fingerprinting-based indoor wireless localization system," *Sensors*, vol. 23, no. 5, art. no. 2545, Feb. 2023, doi: 10.3390/s23052545.

[8] The University of Texas at Austin. (2025, Aug. 25). *Connecting to utexas-iot Wi-Fi* [Online]. Available: https://ut.service-now.com/kb_view.do?sysparm_article=KB0017215.

[9] Espressif Systems. (2026). *ESP-IDF Programming Guide (ESP32-S3): Wi-Fi Vendor Features* [Online]. Available: https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-guides/wifi-driver/wifi-vendor-features.html.

[10] G. Welch and G. Bishop, "An introduction to the Kalman filter," Dept. Comput. Sci., Univ. North Carolina, Chapel Hill, NC, Tech. Rep. TR 95-041, Apr. 2004.

[11] *IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*, IEEE Standard 802.11-2020, Feb. 2021, doi: 10.1109/IEEESTD.2021.9363693.

[12] *Radio Frequency Devices*, 47 C.F.R. Part 15, 2025.

[13] *IEEE Standard for a Smart Transducer Interface for Sensors and Actuators—Common Functions, Communication Protocols, and Transducer Electronic Data Sheet (TEDS) Formats*, IEEE Standard 1451.0-2024, Jun. 2024, doi: 10.1109/IEEESTD.2024.10571828.

[14] R. Chandramouli and E. A. Hibbard, *Guidelines for Media Sanitization*, NIST Special Publication 800-88 Rev. 2, Nat. Inst. Standards Technol., Gaithersburg, MD, Sept. 2025, doi: 10.6028/NIST.SP.800-88r2.

[15] *Security Requirements for Cryptographic Modules*, FIPS PUB 140-3, Nat. Inst. Standards Technol., Gaithersburg, MD, Mar. 2019, doi: 10.6028/NIST.FIPS.140-3.

[16] *Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes—Safety Requirements for Portable Sealed Secondary Cells, and for Batteries Made from Them, for Use in Portable Applications—Part 2: Lithium Systems*, IEC Standard 62133-2:2017, Feb. 2017.

[17] The University of Texas at Austin Information Security Office. (2019, Aug. 28). *Acceptable Use Policy for University Students* [Online]. Available: https://security.utexas.edu/policies/aup.

APPENDIX A – APPLICABLE STANDARDS

The standards and policies in this appendix require BevoTag to use a Wi-Fi chip that exposes CSI and a pre-certified radio module, to separate its sensor drivers from its estimation code, to encrypt its telemetry and erase its memory on reset, to use a certified and protected lithium battery, and to use the campus network only as an ordinary client. Each section below gives the relevant clauses in a table, states their effect on the design, and names the method the team will use to verify compliance.

IEEE 802.11: Wi-Fi Physical and MAC Layers

IEEE 802.11 [11] determines both how BevoTag can measure CSI and how it may join campus networks. Table 6 shows that the physical-layer clauses decide which chips can report CSI, whereas the frame, management, and security clauses decide how the tag listens and authenticates. The team will verify CSI capture by comparing recorded sub-channel data against known access-point beacons, confirm passive listening with a Wireshark capture showing no transmissions during scans, and confirm network association through connection logs on utexas-iot.

Table 6. IEEE 802.11 Requirements Affecting BevoTag

Clause	Requirement	Effect on Design
Clauses 17, 18, and 19 (OFDM, ERP, and HT PHYs)	Define the sub-carrier structure and training fields at the start of each 2.4 GHz packet	The Wi-Fi chip must expose per-sub-carrier CSI from these training fields [9]
Clause 9 (frame formats)	Defines frame headers, including the transmitter address	Firmware reads only header fields, such as the access point address
Clause 11 (MLME)	Defines passive and active scanning and association	Tag scans passively and associates as an ordinary client
Clause 12 (security)	Defines robust security network association, including pre-shared keys	Tag authenticates with the WPA2 pre-shared key that utexas-iot issues [8]

FCC 47 CFR Part 15: Radio Frequency Devices

Part 15 [12] limits both intentional transmissions and unintended emissions from the tag's digital circuits. As Table 7 on the next page shows, a pre-certified Wi-Fi module already satisfies the transmitter and restricted-band limits, which leaves the team responsible mainly for board layout and labeling. The team will verify conducted output power on a bench spectrum analyzer and check the board for unintended emissions with a near-field probe, rather than performing full

certification testing.

Table 7. FCC Part 15 Requirements Affecting BevoTag

Section	Requirement	Effect on Design
§15.109 (unintentional radiators)	Radiated emission limits for Class B digital devices	Processor clocks, sensor buses, and power conversion need careful layout and decoupling
§15.247(b) (intentional radiators)	Conducted output power of at most 1 W (30 dBm) in the 2400–2483.5 MHz band, with antenna gain up to 6 dBi	Use a pre-certified Wi-Fi module whose output stays within this limit
§15.205, §15.209, and §15.247(d) (restricted bands)	Limits on harmonics, band-edge, and spurious emissions	Rely on the certified module's filtering and follow its antenna layout rules
§15.19 and §15.212 (labeling)	Part 15 compliance statement and "Contains FCC ID" label for a modular radio	Enclosure carries the compliance statement and the module's FCC ID
§15.21 (information to user)	Warning that unapproved changes void the user's authority to operate	Documentation includes the required warning

IEEE 1451: Smart Transducer Interface

IEEE 1451.0 [13] guides how the firmware separates sensor hardware from the processing that uses its data. Table 8 shows that the team will adopt the standard's data model and data sheet structure rather than its full command set, which would use more memory than the tag can spare. The team will check compliance through firmware review and unit tests confirming that calibration data load correctly at startup.

Table 8. IEEE 1451.0 Guidelines Affecting BevoTag

Provision	Guideline	Effect on Design
Transducer channel model	Each sensor appears to the processor as a set of standard channels	A hardware abstraction layer separates the IMU and Wi-Fi drivers from the fusion algorithm
Transducer electronic data sheets (TEDS)	Each sensor carries self-describing identification and calibration data	IMU scale factors and bias offsets are stored in a defined format in flash
Common communication functions	Sensor data and status are exchanged through defined message formats	Telemetry packets carry IMU, Wi-Fi, and battery data in a defined format

NIST SP 800-88 and FIPS 140-3: Data Sanitization and Cryptography

These federal guidelines [14], [15] govern how the tag protects and erases the credentials and location history it stores. Table 9 shows that a factory reset will both overwrite stored data and destroy the encryption keys, and that telemetry and firmware will follow FIPS 140-3 practices by relying only on approved algorithms; the team will not seek formal module validation. The team will verify sanitization by reading back flash memory through a debug probe after a factory reset, verify encryption against published NIST test vectors, and verify secure boot by attempting to load an unsigned firmware image.

Table 9. Data Security Requirements Affecting BevoTag

Section	Requirement	Effect on Design
SP 800-88 Clear (Sec. 3.1.1)	Protect against simple recovery through the device's normal interface, for example by a reset to factory state	Factory reset overwrites stored credentials, location history, and survey tables
SP 800-88 Purge by cryptographic erase (Sec. 3.2)	Sanitize the keys that encrypt stored data so the data cannot be decrypted	Reset erases the keys held in the chip's secure storage
FIPS 140-3 approved security functions	Use approved algorithms such as AES and SHA-256	Telemetry encrypted with AES using the chip's hardware accelerator
FIPS 140-3 software and firmware security	Check firmware integrity with an approved digital signature	Secure boot accepts only signed firmware images

IEC 62133-2: Lithium Battery Safety

IEC 62133-2 [16] sets safety requirements for the lithium cell and battery that power the tag. Table 10 on the next page shows that its tests target the same hazards that the Ethical Considerations identify, so the team will buy a cell already certified to the standard and add charge and short-circuit protection on its own board. The team will verify that protection on the bench by checking the charge cutoff voltage, the short-circuit cutoff, and the charge current against the cell's datasheet limits.

UT Austin Acceptable Use Policy

The university's acceptable use policy [17] and its registration rules for utexas-iot [8] govern how the tag uses the campus network. Table 11 on the next page shows that the policy's limits on

Table 10. IEC 62133-2 Requirements Affecting BevoTag

Provision	Requirement	Effect on Design
Overcharge of battery	Battery must not catch fire or explode when charged beyond its limits	Charger limits voltage and current to the cell's rated values
External short circuit	Cell and battery must not catch fire or explode when shorted	Protection circuit disconnects the cell on overcurrent
Thermal abuse	Cell must withstand elevated temperature without fire or explosion	Charging stops outside the cell's rated temperature range
Certified cells	Cells must pass the standard's type tests	Team buys only cells certified to IEC 62133-2

probing and interference match the tag's design as a passive, ordinary network client. The team will verify compliance by reviewing the firmware's network functions against the policy and by confirming with a Wireshark capture that the tag sends only its own telemetry.

Table 11. UT Austin Acceptable Use Policy Provisions Affecting BevoTag

Provision	Requirement	Effect on Design
AUP Sec. 5.10 (scanning)	Never deliberately scan or probe an information resource without authorization	Tag listens passively and never sends probe or scanning traffic
AUP Sec. 5.7 (shared resources)	Do not interfere with others' use or consume a disproportionate share	Tag sends only small, periodic telemetry reports
utexas-iot registration [8]	Register each device's hardware address through the university's network portal	Team registers each tag and disables address randomization